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Topic 4. Continuous time

In this topic

In Topics 1 to 3 a channel use took a number x and returned a noisy number y. A real transmitter sends a waveform, a function of time, and the receiver observes that waveform corrupted by noise. This topic shows that a waveform is a vector, that a finite set of waveforms lives in a space of small dimension, and that projecting the received waveform onto that space gives back the Gaussian channel of Table 1.2 of Topic 1, one use per dimension. The results of the first half of the course for that channel then apply to waveforms without change. Sections 4.1 to 4.3 correspond to the first lecture and Sections 4.4 to 4.6 to the second.

- We define the inner product, the energy and the distance of signals, and see that they obey the rules of vectors.
- We represent signals in an orthonormal basis by their coordinates, and see that energies, inner products and distances are preserved.
- We describe the signals of a transmitter as points of a signal space, and recover the constellations and codes of Topics 1 and 2 as sets of coordinate vectors.
- We define white Gaussian noise by its projections, with variance $N_0/2$ in every direction, and see why its energy is infinite.
- We prove that the part of the noise outside the signal space is irrelevant, and obtain the discrete-time Gaussian channel with $\sigma^2 = N_0/2$.
- We detect in signal space with the nearest-point rule, compute error probabilities from the distance between waveforms, and compare antipodal and orthogonal signaling.

4.1 Signals as vectors

A **signal** is a real function of time $x(t)$ with finite energy, and these are the only signals we consider. In Topic 2 the sequences $\boldsymbol{x} = (x_1, \dots, x_n)$ were treated as vectors: we added them, we computed the squared norm $\|\boldsymbol{x}\|^2 = \sum_{i=1}^{n} x_i^2$ and the squared distance $\|\boldsymbol{y} - \boldsymbol{x}\|^2$, and the ML decoder of Proposition 2.3 of Topic 2 chose the nearest codeword. Signals admit the same operations, with the sum over positions replaced by an integral over time, as sums became integrals in Table 1.1 of Topic 1.

Definition 4.1. The **inner product** of two signals x and y is

$$\langle x, y \rangle = \int_{-\infty}^{\infty} x(t)\, y(t)\, \mathrm{d}t, \tag{4.1}$$

the **energy** of x is $\|x\|^2 = \langle x, x \rangle$, the **norm** $\|x\|$ is the square root of the energy, and the **distance** between x and y is $\|x - y\|$.

The inner product is linear in each argument and symmetric, $\langle x, y \rangle = \langle y, x \rangle$. The energy of ax is $a^2\|x\|^2$, and the energy of a sum is

$$\|x + y\|^2 = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2, \tag{4.2}$$

which follows from expanding the square inside the integral. The squared distance is therefore $\|x - y\|^2 = \|x\|^2 - 2\langle x, y \rangle + \|y\|^2$. A time shift changes none of these quantities: $x(t - \tau)$ has the same energy as $x(t)$, and $\langle x(\cdot - \tau), y(\cdot - \tau) \rangle = \langle x, y \rangle$, because the shift is a change of variable in the integral. Exercise 4.1 shows that $|\langle x, y \rangle| \le \|x\|\, \|y\|$, so the inner product of two unit-energy signals lies between -1 and 1, like the cosine of an angle.

Definition 4.2. Two signals are **orthogonal** when $\langle x, y \rangle = 0$. A set of signals $\{\phi_1, \dots, \phi_n\}$ is **orthonormal** when every signal has unit energy and any two distinct signals are orthogonal, that is, $\langle \phi_i, \phi_k \rangle = 1$ if $i = k$ and 0 otherwise.

Example 4.3. Fix a duration $T > 0$ and consider the two signals of Figure 4.1,

$$\phi_1(t) = \begin{cases} \frac{1}{\sqrt{T}} & 0 \le t < T \\ 0 & \text{otherwise} \end{cases} \qquad \text{and} \qquad \phi_2(t) = \begin{cases} \frac{1}{\sqrt{T}} & 0 \le t < \frac{T}{2} \\ -\frac{1}{\sqrt{T}} & \frac{T}{2} \le t < T \\ 0 & \text{otherwise.} \end{cases} \tag{4.3}$$

Both have unit energy, since ϕ_1^2 and ϕ_2^2 equal $\frac{1}{T}$ on an interval of length T. Their inner product is

$$\langle \phi_1, \phi_2 \rangle = \int_0^{T/2} \frac{1}{T}\,\mathrm{d}t - \int_{T/2}^{T} \frac{1}{T}\,\mathrm{d}t = \frac{1}{2} - \frac{1}{2} = 0, \tag{4.4}$$

so $\{\phi_1, \phi_2\}$ is an orthonormal set. The two signals overlap on the whole interval $[0, T)$, and yet they are orthogonal: the product $\phi_1\phi_2$ is positive on one half and negative on the other, and the two contributions cancel. By (4.2), the signal $a\phi_1 + b\phi_2$ has energy $a^2 + b^2$ for any real a and b. This pair of signals is the running example of the topic.

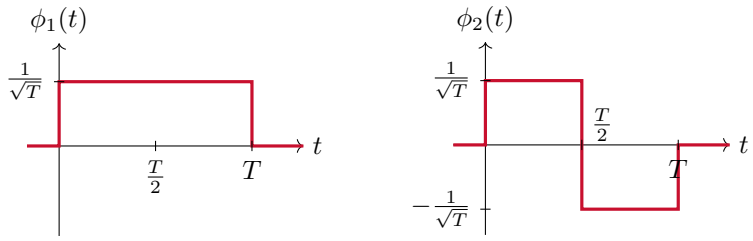


Figure 4.1: The orthonormal set of Example 4.3. The two signals occupy the same interval and are orthogonal.

☞ *Orthogonality is a statement about an integral, not about supports. Two signals with disjoint supports are orthogonal, because their product is zero everywhere, but Example 4.3 shows that orthogonal signals may overlap completely. This is what will allow Topic 5 to send many symbols at once in a limited bandwidth.*

4.2 Orthonormal bases and coordinates

Given an orthonormal set $\{\phi_1, \ldots, \phi_n\}$, the signals that can be written as

$$x(t) = \sum_{i=1}^{n} x_i\, \phi_i(t), \tag{4.5}$$

with real **coordinates** $x_1, \ldots, x_n$, form the **signal space** spanned by the set, and the set is an **orthonormal basis** of that space. Its **dimension** is n. The next result says that such a signal is completely described by its n coordinates, and that the vector operations of Topic 2 on the coordinates give the signal operations of Section 4.1.

Proposition 4.4. Let x and y be two signals of the space spanned by the orthonormal basis $\{\phi_1, \ldots, \phi_n\}$, with coordinates $\boldsymbol{x} = (x_1, \ldots, x_n)$ and $\boldsymbol{y} = (y_1, \ldots, y_n)$. Then,

(a) the coordinates are recovered by inner products, $x_i = \langle x, \phi_i \rangle$,
(b) the inner product of the signals is that of the coordinate vectors, $\langle x, y \rangle = \sum_{i=1}^{n} x_i y_i$,
(c) the energy of the signal is $\|x\|^2 = \sum_{i=1}^{n} x_i^2 = \|\boldsymbol{x}\|^2$, and
(d) the distance between the signals is that between the vectors, $\|x - y\|^2 = \sum_{i=1}^{n} (x_i - y_i)^2 = \|\boldsymbol{x} - \boldsymbol{y}\|^2$.

Proof. By the linearity of the inner product and (4.5), $\langle x, \phi_i \rangle = \sum_{k=1}^{n} x_k \langle \phi_k, \phi_i \rangle$, and the only nonzero term of the sum is $k = i$, which gives x_i. For (b), we expand $\langle x, y \rangle = \sum_{i=1}^{n} \sum_{k=1}^{n} x_i y_k \langle \phi_i, \phi_k \rangle$ and keep the terms with $i = k$. Then (c) is (b) with $y = x$, and (d) is (c) applied to the signal $x - y$, whose coordinates are $x_i - y_i$. □

Problem 4.5. The signal $x(t)$ equals 3 for $0 \le t < \frac{T}{2}$, equals 1 for $\frac{T}{2} \le t < T$ and is zero otherwise. Find its coordinates in the basis of Example 4.3 and check Proposition 4.4(c).

Solution: By Proposition 4.4(a), the coordinates are the two inner products

$$x_1 = \langle x, \phi_1 \rangle = \frac{1}{\sqrt{T}}\Big(3 \cdot \frac{T}{2} + 1 \cdot \frac{T}{2}\Big) = 2\sqrt{T} \quad \text{and} \quad x_2 = \langle x, \phi_2 \rangle = \frac{1}{\sqrt{T}}\Big(3 \cdot \frac{T}{2} - 1 \cdot \frac{T}{2}\Big) = \sqrt{T}, \tag{4.6}$$

and indeed $2\sqrt{T}\phi_1(t) + \sqrt{T}\phi_2(t)$ equals $2 + 1 = 3$ on the first half and $2 - 1 = 1$ on the second. The energy computed from the signal is $\int_0^{T/2} 9\,\mathrm{d}t + \int_{T/2}^{T} 1\,\mathrm{d}t = 5T$, and from the coordinates it is $x_1^2 + x_2^2 = 4T + T = 5T$. The first coordinate is the average level of the signal, scaled by $\sqrt{T}$, and the second is half the jump between the two halves, scaled the same way.

A signal that is not in the space can still be compared with it. For any signal x, the signal

$$\bar{x}(t) = \sum_{i=1}^{n} \langle x, \phi_i \rangle \, \phi_i(t) \tag{4.7}$$

is its **projection** onto the space, and $e = x - \bar{x}$ is the **error** of the projection.

Proposition 4.6. The error e is orthogonal to every signal of the space, and $\|x\|^2 = \|\bar{x}\|^2 + \|e\|^2$. The projection is the signal of the space nearest to x: for every signal y of the space, $\|x - y\|^2 = \|e\|^2 + \|\bar{x} - y\|^2$.

Proof. By construction $\langle e, \phi_i \rangle = \langle x, \phi_i \rangle - \langle \bar{x}, \phi_i \rangle = 0$ for every i, by Proposition 4.4(a) applied to $\bar{x}$, so e is orthogonal to every combination of the ϕ_i, in particular to $\bar{x}$ and to $\bar{x} - y$. Then (4.2) with $x = \bar{x} + e$ gives the first identity, and with $x - y = (\bar{x} - y) + e$ the second. □

Exercise 4.2 projects a ramp onto the basis of Example 4.3. In the other direction, any finite set of signals spans a space with an orthonormal basis, which Exercise 4.3 builds for two signals. The procedure, called Gram-Schmidt, starts with the first nonzero signal and normalizes it. For each remaining signal it subtracts the projection onto the span of the basis signals already obtained, discards the error if it is zero, and otherwise normalizes it to obtain the next basis signal. We use it only to know that the basis exists.

4.3 The signal space of a transmitter

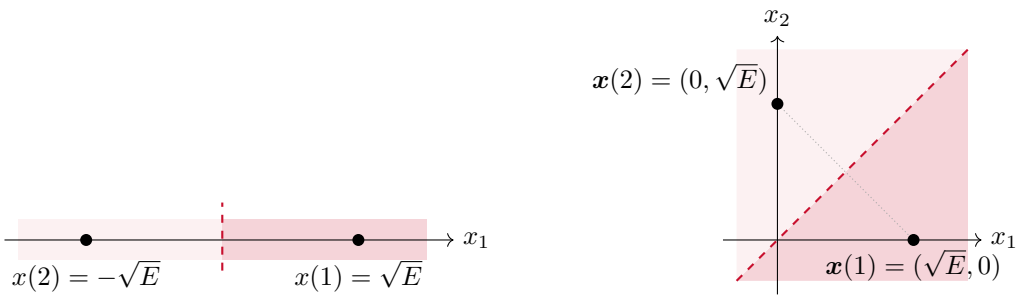
A transmitter has $M = 2^k$ messages $m \in \mathcal{M}$, with prior probabilities $\Pr(m)$, and sends the waveform $s_m(t)$ for message m. The M waveforms span a signal space of dimension $n \le M$, with an orthonormal basis $\{\phi_1, \ldots, \phi_n\}$, so by Proposition 4.4 each of them is

$$s_m(t) = \sum_{i=1}^{n} x_i(m)\, \phi_i(t), \qquad x_i(m) = \langle s_m, \phi_i \rangle, \tag{4.8}$$

and is described by its coordinate vector $\boldsymbol{x}(m) = (x_1(m), \ldots, x_n(m))$. These vectors are precisely the codewords of Definition 2.1 of Topic 2, and when $n = 1$ they are the symbols of a constellation of Topic 1. The **energy** of message m is $E_m = \|s_m\|^2 = \|\boldsymbol{x}(m)\|^2$, by Proposition 4.4(c), the average energy is $E = \sum_{m \in \mathcal{M}} \Pr(m) E_m$, and the energy per information bit is $E_\mathrm{b} = E/k$, as in Topic 2.

The transmitter is thus a bank of n multipliers and an adder, as the left part of Figure 4.4 shows: it scales each basis waveform by a coordinate and adds the results. The simplest basis is a unit-energy pulse $\phi(t)$ supported on $[0, T)$ and its shifts $\phi_i(t) = \phi\big(t - (i-1)T\big)$, which are orthonormal because their supports are disjoint. Sending the symbols $x_1, \ldots, x_n$ of Topics 1 and 2 with this basis means sending $x_i \phi(t - (i-1)T)$ in the i-th slot of T seconds, one coordinate per slot. Topic 5 builds bases of shifted pulses whose supports overlap, as ϕ_1 and ϕ_2 do, and asks how little bandwidth they need.

Example 4.7. Two messages, $M = 2$, are sent with waveforms of the same energy E built on the basis of Example 4.3. In **antipodal signaling**, $s_1 = \sqrt{E}\,\phi_1$ and $s_2 = -\sqrt{E}\,\phi_1$: the dimension is $n = 1$, the coordinates are $x(1) = \sqrt{E}$ and $x(2) = -\sqrt{E}$, and the distance between the two waveforms is $2\sqrt{E}$. This is the 2-PAM of Example 1.6 of Topic 1 with $A = \sqrt{E}$, now as two rectangular pulses of opposite sign. In **orthogonal signaling**, $s_1 = \sqrt{E}\,\phi_1$ and $s_2 = \sqrt{E}\,\phi_2$: the dimension is $n = 2$, the coordinates are $\boldsymbol{x}(1) = (\sqrt{E}, 0)$ and $\boldsymbol{x}(2) = (0, \sqrt{E})$, and the squared distance is $\|\boldsymbol{x}(1) - \boldsymbol{x}(2)\|^2 = E + E = 2E$. Figure 4.2 shows the two configurations. At equal energy, the antipodal pair is farther apart, by a factor $\sqrt{2}$ in distance. Section 4.6 turns this factor into 3 dB.



(a) antipodal, $n = 1$ (b) orthogonal, $n = 2$

Figure 4.2: The two binary signal sets of Example 4.7, with the decision regions of the nearest-point detector of Section 4.6. The dotted segment has length $\sqrt{2E}$, against $2\sqrt{E}$ in (a).

Problem 4.8. Four equiprobable messages are sent with the waveforms $s_1 = a\phi_1 + a\phi_2$, $s_2 = -a\phi_1 + a\phi_2$, $s_3 = -a\phi_1 - a\phi_2$ and $s_4 = a\phi_1 - a\phi_2$, with $a > 0$ and the basis of Example 4.3. Draw the four waveforms, and find the energy of each message, the energy per bit and the smallest distance between two waveforms.

Solution: On the first half of the interval ϕ_1 and ϕ_2 both equal $\frac{1}{\sqrt{T}}$, and on the second half they have opposite signs. Hence, s_1 equals $\frac{2a}{\sqrt{T}}$ on $[0, \frac{T}{2})$ and 0 on $[\frac{T}{2}, T)$, s_2 equals 0 and then $-\frac{2a}{\sqrt{T}}$, s_3 equals $-\frac{2a}{\sqrt{T}}$ and then 0, and s_4 equals 0 and then $\frac{2a}{\sqrt{T}}$, as in Figure 4.3. Each message is a pulse of amplitude $\pm\frac{2a}{\sqrt{T}}$ placed in the first or in the second half: the message is carried by the position and by the sign of the pulse. By Proposition 4.4(c), every message has energy $a^2 + a^2 = 2a^2$, which is also $\frac{4a^2}{T} \cdot \frac{T}{2}$ computed from the waveform, so $E = 2a^2$ and, with $k = 2$ bits, $E_\mathrm{b} = a^2$. The coordinate vectors are the four corners $(\pm a, \pm a)$ of a square, the real version of the QPSK constellation of Problem 1.13 of Topic 1. Two neighboring corners differ in one coordinate by $2a$, so the smallest squared distance is $4a^2$, and the diagonal pairs are at squared distance $8a^2$. In terms of the energy per bit, the minimum squared distance is $4E_\mathrm{b}$, the same as for antipodal signaling, where $d^2 = 4E = 4E_\mathrm{b}$. The four signals carry two bits in two dimensions at no cost in distance per bit.

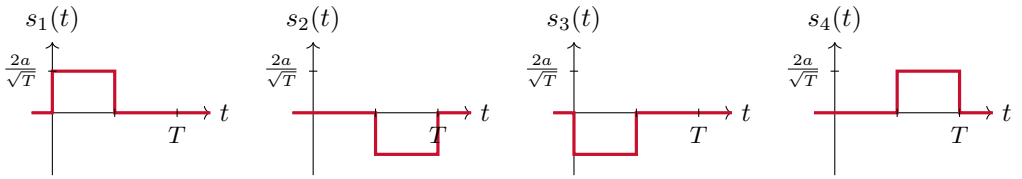


Figure 4.3: The four waveforms of Problem 4.8: a pulse in the first or the second half of the interval, with either sign. Their coordinate vectors are the corners $(\pm a, \pm a)$.

The codes of Topic 2 fit in the same picture. The parity code of Problem 2.9 of Topic 2 sends three 2-PAM symbols, so with three slots it has $n = 3$ and its four codewords are the vertices of the cube of Figure 2.4 of that topic. The geometry that governed the comparison of codes, minimum distance at equal energy per bit, is the geometry of the signal space, and it does not depend on which orthonormal waveforms carry the coordinates.

4.4 White Gaussian noise

The channel adds to the transmitted waveform a noise waveform $Z(t)$, which does not depend on the message, and the receiver observes $Y(t) = s_m(t) + Z(t)$. Everything the receiver can compute from $Z(t)$ alone is independent of m. The noise of the first half of the course was a sequence of independent Gaussian samples. Its continuous-time counterpart is defined not by its values at each instant, which we will never use, but by what the receiver can compute from it, namely inner products with signals.

Definition 4.9. The noise Z is **white Gaussian noise** with **power spectral density** $N_0/2$ when, for every finite collection of signals $u_1, \dots, u_J$, the random variables $\langle Z, u_1 \rangle, \dots, \langle Z, u_J \rangle$ are jointly Gaussian with zero mean, and for every pair of signals u and v,

$$\mathrm{E}\big(\langle Z, u \rangle \, \langle Z, v \rangle\big) = \frac{N_0}{2} \, \langle u, v \rangle. \tag{4.9}$$

The expressions $Z(t)$ and $\langle Z, u\rangle$ are formal notation for this family of Gaussian projections, not for pointwise values of an ordinary random function. The definition is the mathematical form of two physical ideas. The first is that the noise observed in one interval of time says nothing about the noise in another: two signals with disjoint supports have $\langle u, v\rangle = 0$, so by (4.9) their projections are uncorrelated and, being jointly Gaussian, independent. The second is about intensity. Taking $v = u$ in (4.9), the variance of $\langle Z, u\rangle$ is $\frac{N_0}{2}\|u\|^2$. It depends on the energy of u and on nothing else, not on its duration, its position in time or its shape, and this is what the word white means: the noise is seen with the same intensity through every unit-energy signal, as white light contains every color equally. The constant $N_0/2$ is measured in watts per hertz, which is the same as joules, and Topic 6 relates it to the bandwidth of the receiver. With the convention of the factor $\frac{1}{2}$, the noise power in a band of W hertz is $N_0 W$.

Proposition 4.10. Let $\{\phi_1, \dots, \phi_n\}$ be an orthonormal set. The projections $Z_i = \langle Z, \phi_i\rangle$ of white Gaussian noise are independent Gaussian random variables with zero mean and variance $\sigma^2 = N_0/2$.

Proof. By (4.9), $\mathrm{E}(Z_i^2) = \frac{N_0}{2}\|\phi_i\|^2 = \frac{N_0}{2}$ and $\mathrm{E}(Z_i Z_k) = \frac{N_0}{2}\langle \phi_i, \phi_k\rangle = 0$ for $i \neq k$. The projections are jointly Gaussian and uncorrelated, and uncorrelated jointly Gaussian variables are independent, as seen in the probability course. □

Example 4.11. With ϕ_1 of Example 4.3, the projection $Z_1 = \frac{1}{\sqrt{T}}\int_0^T Z(t)\,\mathrm{d}t$ is the integral of the noise over the interval divided by $\sqrt{T}$, and it is Gaussian with variance $N_0/2$ whatever the value of T. A rectangular pulse of amplitude A on the same interval has coordinate $\langle A\sqrt{T}\phi_1, \phi_1\rangle = A\sqrt{T}$ along ϕ_1. Integrating over a longer interval and dividing by $\sqrt{T}$ gives the same noise variance but a coordinate that grows like $\sqrt{T}$, so the ratio of the squared coordinate to the noise variance, $2A^2T/N_0$, grows linearly with the duration of the pulse. Energy, not amplitude, is what buys reliability.

☞ *White noise is not a signal. If it were, by Proposition 4.6 its energy would be at least the sum $Z_1^2 + \cdots + Z_K^2$ of its squared projections onto any K orthonormal signals, for every K. Along an infinite orthonormal sequence, Proposition 4.10 gives independent, identically distributed Gaussian projections. By the strong law of large numbers, the average $(Z_1^2 + \cdots + Z_K^2)/K$ converges almost surely to $N_0/2 > 0$, so the sum diverges almost surely. No finite signal energy can bound these sums. The model is an idealization of a noise whose bandwidth is much larger than that of the signals, and it is harmless because, as the next section shows, only n projections of the noise ever reach the receiver's decision.*

4.5 Irrelevance and the vector channel

The receiver observes $Y(t) = s_m(t) + Z(t)$ and may compute any finite number of inner products of Y with signals of its choice. The natural choice is the basis of the transmitter. By (4.8) and the linearity of the inner product, the projection of the received waveform onto ϕ_i is

$$Y_i = \langle Y, \phi_i\rangle = \langle s_m, \phi_i\rangle + \langle Z, \phi_i\rangle = x_i(m) + Z_i, \qquad i = 1, \dots, n, \tag{4.10}$$

the coordinate of the transmitted waveform plus the projection of the noise. The right part of Figure 4.4 shows the n **correlators** that compute them. The question is whether anything is lost by keeping only these n numbers out of a whole waveform. The following theorem says that nothing is.

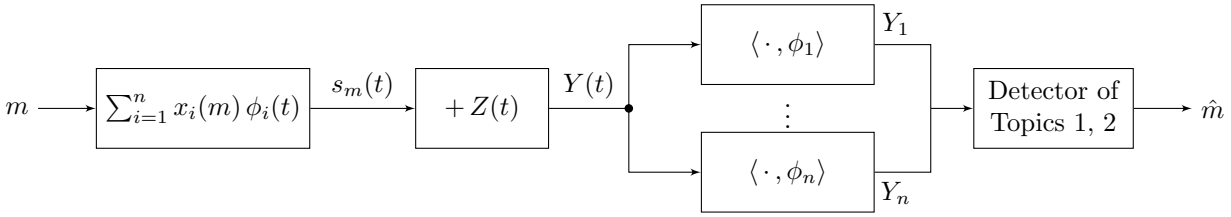


Figure 4.4: Transmission of waveforms in signal space. The transmitter combines the basis waveforms with the coordinates of the message, and the receiver recovers the noisy coordinates with a bank of correlators and decides as in Topics 1 and 2.

Theorem 4.12 (Irrelevance)**.** Let $\psi_1, \dots, \psi_J$ be orthonormal signals, each orthogonal to $\phi_1, \dots, \phi_n$, and let $V_j = \langle Y, \psi_j\rangle$. The random variables $V_1, \dots, V_J$ do not depend on the message and are independent of $Z_1, \dots, Z_n$.

Consequently, the MAP decision based on $(Y_1, \dots, Y_n, V_1, \dots, V_J)$ is the MAP decision based on $(Y_1, \dots, Y_n)$ alone.

Proof. By (4.8), $\langle s_m, \psi_j \rangle = \sum_{i=1}^{n} x_i(m) \langle \phi_i, \psi_j \rangle = 0$, so $V_j = \langle Z, \psi_j \rangle$ is a projection of the noise alone and does not depend on m. By (4.9), $\mathrm{E}(V_j Z_i) = \frac{N_0}{2} \langle \psi_j, \phi_i \rangle = 0$, and jointly Gaussian uncorrelated variables are independent, so the vector $\boldsymbol{V} = (V_1, \dots, V_J)$ is independent of $(Z_1, \dots, Z_n)$, and therefore of $\boldsymbol{Y} = (Y_1, \dots, Y_n)$ given m. By Proposition 4.10 applied to the orthonormal set $\{\psi_j\}$, the V_j are independent Gaussians of variance $N_0/2$, so $\boldsymbol{V}$ has a density f. The joint density of the observation given m is then a product $W(\boldsymbol{y}|\boldsymbol{x}(m))\, f(\boldsymbol{v})$, where f does not depend on m. The MAP decoder of Theorem 2.2 of Topic 2 maximizes $\Pr(m)\, W(\boldsymbol{y}|\boldsymbol{x}(m))\, f(\boldsymbol{v})$ over m, and the factor $f(\boldsymbol{v})$ is common to all messages and does not affect the maximization. □

Any signal ψ is the sum of its projection onto the signal space and an error orthogonal to it, by Proposition 4.6. If the error is nonzero, $\langle Y, \psi \rangle$ is a combination of $Y_1, \dots, Y_n$ and of the projection onto the normalized error, a variable V covered by the theorem. If the error is zero, the observation is already a combination of $Y_1, \dots, Y_n$. The n correlator outputs are therefore all that the optimal receiver needs. Together with Proposition 4.10, equation (4.10) says that

$$\boldsymbol{Y} = \boldsymbol{x}(m) + \boldsymbol{Z}, \qquad Z_1, \dots, Z_n \text{ independent, Gaussian, of variance } \sigma^2 = \frac{N_0}{2}. \tag{4.11}$$

This is the real Gaussian channel of Table 1.2 of Topic 1, used n times, once per dimension of the signal space, and memoryless because the noise projections are independent. A **channel use**, the abstract unit of the first three topics, is one orthonormal direction of the signal space. Table 4.1 collects the correspondence. In particular, for a transmitter with equal-energy waveforms, the energy per channel use is $E_\mathrm{s} = E/n$ and the signal-to-noise ratio of Topic 1 reads

$$\mathrm{SNR} = \frac{E_\mathrm{s}}{\sigma^2} = \frac{2E_\mathrm{s}}{N_0}. \tag{4.12}$$

Topics 1 to 3	This topic
Channel use	One orthonormal waveform ϕ_i, one dimension of the signal space
Symbol x_i, codeword $\boldsymbol{x}(m)$	Coordinate $x_i(m) = \langle s_m, \phi_i \rangle$, waveform $s_m(t)$
Received symbol y_i	Correlator output $Y_i = \langle Y, \phi_i \rangle$
Noise sample of variance σ^2	Noise projection $\langle Z, \phi_i \rangle$ of variance $N_0/2$
Squared distance $\|\boldsymbol{x}(m) - \boldsymbol{x}(m')\|^2$	Squared distance between waveforms, $\|s_m - s_{m'}\|^2 = \|\boldsymbol{x}(m) - \boldsymbol{x}(m')\|^2$
Energy E_s per channel use, $\mathrm{SNR} = E_\mathrm{s}/\sigma^2$	Energy E/n per dimension, $\mathrm{SNR} = 2E_\mathrm{s}/N_0$

Table 4.1: The dictionary between the discrete-time channel of Topics 1 to 3 and the waveform channel.

☞ *Topic 1 took the noise variance σ^2 as given, and the complex Gaussian channel of Table 1.2 had total variance σ^2 with $\sigma^2/2$ per real part. Both conventions come from here. A real coordinate carries noise of variance $N_0/2$, and Topic 5 pairs two real coordinates, carried by a cosine and a sine, into one complex channel use whose noise has total variance $\sigma^2 = N_0$, that is, $N_0/2$ per real component.*

4.6 Detection in signal space

With (4.11), the results of Topics 1 and 2 apply to waveforms at once. The MAP decoder is optimal by Theorem 2.2 of Topic 2 and, for equiprobable messages, it is the ML decoder, which by Proposition 2.3(a) of that topic chooses the coordinate vector nearest to $\boldsymbol{Y}$. By Proposition 4.4(d), this is the waveform s_m nearest to the projection $\sum_{i=1}^{n} Y_i \phi_i(t)$ of the received waveform onto the signal space. The decision regions are the Voronoi regions of the points $\boldsymbol{x}(m)$, as in Section 1.5 of Topic 1 and in Figure 4.2. The rule can also be written without the basis.

Proposition 4.13. The ML decoder is

$$\hat{m} = \arg\min_{m \in \mathcal{M}} \|\boldsymbol{Y} - \boldsymbol{x}(m)\|^2 = \arg\max_{m \in \mathcal{M}} \Big(\langle Y, s_m \rangle - \frac{E_m}{2} \Big), \tag{4.13}$$

and when all the waveforms have the same energy it chooses the waveform most correlated with the received one. For equiprobable messages, this is also the MAP decoder.

Proof. Expanding the square, $\|\boldsymbol{Y}-\boldsymbol{x}(m)\|^2 = \|\boldsymbol{Y}\|^2 - 2\sum_{i=1}^{n} Y_i x_i(m) + \|\boldsymbol{x}(m)\|^2$. The first term does not depend on m, the last one is E_m, and by (4.8) and linearity $\sum_{i=1}^{n} x_i(m)\langle Y, \phi_i\rangle = \langle Y, s_m\rangle$. Minimizing the distance is thus maximizing $\langle Y, s_m\rangle - E_m/2$, and the energy term drops out when it is the same for all messages. □

The receiver may thus correlate the received waveform with the M candidate waveforms instead of the n basis waveforms. The two implementations give the same decision. Topic 5 shows that a correlation can be computed by filtering and sampling, which is how receivers are built.

For error probabilities we need no new computation either. For two messages at distance $d = \|\boldsymbol{x}(1)-\boldsymbol{x}(2)\| = \|s_1 - s_2\|$, the ML decoder finds the second more likely than the first with probability $\frac{1}{2}\operatorname{erfc}\big(d/(2\sqrt{2}\,\sigma)\big)$, by Exercise 2.7 of Topic 2, and for $M = 2$ this is exactly the error probability. With $\sigma^2 = N_0/2$, we have $2\sqrt{2}\,\sigma = 2\sqrt{N_0}$ and

$$P_{\mathrm{e}} = \tfrac{1}{2}\operatorname{erfc}\left(\frac{d}{2\sqrt{N_0}}\right). \tag{4.14}$$

The error probability of binary signaling in white noise depends on the two waveforms only through their distance.

Example 4.14. For the two signal sets of Example 4.7, equation (4.14) gives, with $d = 2\sqrt{E}$ and $d = \sqrt{2E}$,

$$P_{\mathrm{e}}^{\text{antipodal}} = \tfrac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{E}{N_0}}\right) \qquad \text{and} \qquad P_{\mathrm{e}}^{\text{orthogonal}} = \tfrac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{E}{2N_0}}\right). \tag{4.15}$$

Orthogonal signaling needs twice the energy of antipodal signaling, that is, 3 dB more, for the same error probability. This is the factor $\sqrt{2}$ in distance of Example 4.7, squared. Since each message carries one bit, $E = E_{\mathrm{b}}$ in both cases. As a check, antipodal signaling is 2-PAM with $A = \sqrt{E}$ on a channel of variance $N_0/2$, so $\mathrm{SNR} = 2E/N_0$ by (4.12), and equation (1.22) of Topic 1 gives $\frac{1}{2}\operatorname{erfc}(\sqrt{\mathrm{SNR}/2}) = \frac{1}{2}\operatorname{erfc}(\sqrt{E/N_0})$, the same value. Table 4.2 summarizes the comparison.

Problem 4.15. Find the decision regions and the error probability of the ML decoder for the four waveforms of Problem 4.8, and compare with antipodal signaling at the same energy per bit.

Solution: The coordinate vectors are the corners $(\pm a, \pm a)$ and the nearest-point regions are the four quadrants of the plane (Y_1, Y_2), as for the QPSK of Problem 1.13 of Topic 1. When s_1 is sent, $\boldsymbol{Y} = (a+Z_1, a+Z_2)$ stays in the first quadrant when $Z_1 > -a$ and $Z_2 > -a$. The two conditions concern independent Gaussians of variance $N_0/2$, so by the product rule and equation (1.23) of Topic 1 with $\sigma^2 = N_0/2$,

$$\Pr(\text{correct} \mid m = 1) = (1-p)^2, \qquad p = \Pr(Z_1 < -a) = \tfrac{1}{2}\operatorname{erfc}\left(\frac{a}{\sqrt{N_0}}\right), \tag{4.16}$$

and $P_{\mathrm{e}}(1) = 1 - (1-p)^2 = 2p - p^2$. By symmetry this is the error probability of every message, and hence the average. Since $E_{\mathrm{b}} = a^2$ by Problem 4.8, $p = \frac{1}{2}\operatorname{erfc}\big(\sqrt{E_{\mathrm{b}}/N_0}\big)$. If the two bits of each message are assigned to the signs of the two coordinates, each bit is decided from its own coordinate and is wrong with probability p exactly, as in Exercise 1.5 of Topic 1, so the bit error probability is $P_{\mathrm{b}} = p = \frac{1}{2}\operatorname{erfc}\big(\sqrt{E_{\mathrm{b}}/N_0}\big)$, the same as for antipodal signaling in (4.15). The four waveforms are two independent antipodal transmissions, one per dimension, and cost the same energy per bit as one. The comparison with 4-PAM in one dimension, Problem 1.12 of Topic 1, is instructive: 4-PAM needs five times the symbol energy of 2-PAM for about the same error probability, which is 7 dB per symbol and, since each of its symbols carries two bits, about 4 dB per bit, whereas a second dimension carries the second bit at no cost per bit. Dimensions are a resource, and Topic 6 says what they cost in bandwidth.

Signal set	n	bits	E	$d_{\min}^2/E_{\mathrm{b}}$	P_{e}
Antipodal, $\pm\sqrt{E}\,\phi_1$	1	1	E_{b}	4	$\frac{1}{2}\operatorname{erfc}\big(\sqrt{E_{\mathrm{b}}/N_0}\big)$
Orthogonal, $\sqrt{E}\,\phi_1$ and $\sqrt{E}\,\phi_2$	2	1	E_{b}	2	$\frac{1}{2}\operatorname{erfc}\big(\sqrt{E_{\mathrm{b}}/(2N_0)}\big)$
Four corners $(\pm a, \pm a)$	2	2	$2E_{\mathrm{b}}$	4	$2p - p^2$, $p = \frac{1}{2}\operatorname{erfc}\big(\sqrt{E_{\mathrm{b}}/N_0}\big)$

Table 4.2: The three signal sets of this section on the basis of Example 4.3, at the same energy per bit.

☞ *Nothing in this section used the shape of ϕ_1 and ϕ_2. Any orthonormal pair, rectangular pulses, sinusoids or the pulses of Topic 5, gives the same coordinates, the same distances and the same error probabilities. The shape of the waveforms decides how much time and bandwidth they occupy, which is the subject of the next two topics, not how reliable they are.*

What to remember

- Signals are vectors: $\langle x, y\rangle = \int x(t)y(t)\,\mathrm{d}t$ is their inner product, $\|x\|^2 = \langle x, x\rangle$ their energy and $\|x - y\|$ their distance. Orthogonal signals have zero inner product and need not have disjoint supports.
- In an orthonormal basis, a signal of the spanned space is its coordinates $x_i = \langle x, \phi_i\rangle$, and inner products, energies and distances of signals are those of the coordinate vectors. The projection $\bar{x} = \sum_i \langle x, \phi_i\rangle \phi_i$ is the nearest signal of the space, with an orthogonal error.
- The M waveforms of a transmitter span a space of dimension $n \leq M$. Their coordinate vectors $\boldsymbol{x}(m)$ are the codewords of Topic 2, and a channel use is one dimension.
- White Gaussian noise of spectral density $N_0/2$ has, in every orthonormal direction, an independent Gaussian projection of variance $N_0/2$, and infinite energy. The projections outside the signal space are irrelevant to the optimal decision.
- Correlating the received waveform with the basis gives $\boldsymbol{Y} = \boldsymbol{x}(m) + \boldsymbol{Z}$, the real Gaussian channel of Topic 1 with $\sigma^2 = N_0/2$, used n times. The results of Topics 1 to 3 for that channel apply, with $\mathrm{SNR} = 2E_\mathrm{s}/N_0$.
- In binary signaling under ML detection, two waveforms at distance d are confused with probability $\frac{1}{2}\operatorname{erfc}\big(d/(2\sqrt{N_0})\big)$. Antipodal signaling gives $\frac{1}{2}\operatorname{erfc}(\sqrt{E_\mathrm{b}/N_0})$ and orthogonal signaling needs 3 dB more. A second dimension carries a second bit at the same energy per bit.

Exercises

The last two exercises are more demanding than the others.

Exercise 4.1. Let x and y be two signals with $\|y\| > 0$, and let $a = \langle x, y\rangle / \|y\|^2$. Compute the energy of $x - ay$ with (4.2) and deduce the inequality $|\langle x, y\rangle| \leq \|x\|\,\|y\|$. When does equality hold?

Exercise 4.2. The ramp $x(t) = t$ for $0 \leq t < T$, and 0 otherwise, is projected onto the basis of Example 4.3. Find its coordinates, draw the projection $\bar{x}$, and compute the energies of x, of $\bar{x}$ and of the error $x - \bar{x}$. Check Proposition 4.6.

Exercise 4.3. Let $s_1(t) = 1$ for $0 \leq t < T$ and $s_2(t) = 1$ for $0 \leq t < \frac{T}{2}$, both zero otherwise. Build an orthonormal basis of the space they span: normalize s_1, subtract from s_2 its projection onto the result, and normalize the error. Compare with Example 4.3, and give the coordinates of s_2 in the basis obtained.

Exercise 4.4. The antipodal waveforms $\pm\sqrt{E}\,\phi_1$ of Example 4.7 are received in white noise, but the receiver correlates Y with the unit-energy signal $u(t) = \sqrt{2/T}$ for $0 \leq t < \frac{T}{2}$, zero otherwise, and decides from the sign of $\langle Y, u\rangle$. Find the law of $\langle Y, u\rangle$ under each message and the error probability, and compare with (4.15). What happens with a signal u orthogonal to ϕ_1?

Exercise 4.5. The orthogonal waveforms of Example 4.7 are sent with $\Pr(m = 1) = q$. Show that the MAP decoder decides $m = 1$ when $Y_1 - Y_2$ exceeds a threshold, find it, and identify the result with Problem 1.7 of Topic 1 for a suitable amplitude and noise variance. Compute the error probability for $q = \frac{1}{2}$ and check (4.15).

Exercise 4.6. A pulse $\pm\sqrt{E}\,\phi_1(t)$ is repeated in n consecutive slots, $s_1(t) = \sqrt{E}\sum_{i=1}^{n}\phi_1\big(t - (i-1)T\big)$ and $s_2 = -s_1$. Give an orthonormal basis, the coordinate vectors and the ML decoder, and show from Example 2.4 and equation (2.15) of Topic 2 that $P_\mathrm{e} = \frac{1}{2}\operatorname{erfc}\big(\sqrt{nE/N_0}\big)$. Compare with a single pulse of amplitude $\pm\sqrt{E/T}$ and duration nT.

Exercise 4.7. Two equiprobable messages are sent with $s_1(t) = A$ and $s_2(t) = A(1 - 2t/T)$ for $0 \leq t < T$, both zero otherwise. Without finding a basis, compute $\langle s_1, s_2\rangle$, the two energies, the distance between the waveforms and the error probability of the ML decoder in white noise of spectral density $N_0/2$. Compare with antipodal signaling of the same average energy.

Exercise 4.8. Let M equiprobable messages be sent with M orthogonal waveforms of energy E, for instance a pulse in one of M consecutive slots. Give the coordinate vectors and the distance between any two of them. Use Exercise 2.7 of Topic 2 and the sum of the pairwise probabilities to show that $P_\mathrm{e} \leq \frac{M-1}{2}\operatorname{erfc}\big(\sqrt{E/(2N_0)}\big)$, and write the bound in terms of E_b. Evaluate it at $E_\mathrm{b}/N_0 = 4$ for $M = 2$, 4 and 16 with Table 1.3 of Topic 1 and $\operatorname{erfc}(2\sqrt{2}) \approx 6.3 \cdot 10^{-5}$, and compare the trend with that of M-PAM in Exercise 1.2 of Topic 1.

Solutions

Exercise 4.1. By (4.2) with $-ay$ in the place of y, the energy of $x - ay$ is $\|x\|^2 - 2a\langle x, y\rangle + a^2\|y\|^2$. With $a = \langle x, y\rangle / \|y\|^2$ the last two terms are $-2\langle x, y\rangle^2/\|y\|^2 + \langle x, y\rangle^2/\|y\|^2$, so

$$\|x - ay\|^2 = \|x\|^2 - \frac{\langle x, y\rangle^2}{\|y\|^2}. \tag{4.17}$$

An energy is nonnegative, so $\langle x, y\rangle^2 \le \|x\|^2\|y\|^2$, which is the inequality after taking square roots. Equality holds when $x - ay$ has zero energy, that is, when $x = ay$ is a multiple of y. The quantity ay is the projection of x onto the one-dimensional space spanned by y, and the identity is Proposition 4.6 for $n = 1$: the energy of x is that of its projection plus that of the error. For two unit-energy signals the inner product lies in $[-1, 1]$, and it equals ± 1 only when the signals are equal or opposite.

Exercise 4.2. The coordinates are the inner products with ϕ_1 and ϕ_2,

$$x_1 = \frac{1}{\sqrt{T}}\int_0^T t\,\mathrm{d}t = \frac{T^{3/2}}{2} \qquad \text{and} \qquad x_2 = \frac{1}{\sqrt{T}}\Big(\int_0^{T/2} t\,\mathrm{d}t - \int_{T/2}^T t\,\mathrm{d}t\Big) = \frac{1}{\sqrt{T}}\Big(\frac{T^2}{8} - \frac{3T^2}{8}\Big) = -\frac{T^{3/2}}{4}. \tag{4.18}$$

The projection $\bar{x} = x_1\phi_1 + x_2\phi_2$ equals $\frac{T}{2} - \frac{T}{4} = \frac{T}{4}$ on the first half and $\frac{T}{2} + \frac{T}{4} = \frac{3T}{4}$ on the second, as Figure 4.5 shows: the basis replaces the ramp by its average on each half of the interval. The energies are $\|x\|^2 = \int_0^T t^2\,\mathrm{d}t = \frac{T^3}{3}$ and $\|\bar{x}\|^2 = x_1^2 + x_2^2 = \frac{T^3}{4} + \frac{T^3}{16} = \frac{5T^3}{16}$, so the error has energy $\frac{T^3}{3} - \frac{5T^3}{16} = \frac{T^3}{48}$, which is the energy of $x - \bar{x}$ computed directly: the error is $t - \frac{T}{4}$ on the first half and $t - \frac{3T}{4}$ on the second, two ramps of slope 1 centered on intervals of length $\frac{T}{2}$, each of energy $\int_{-T/4}^{T/4} u^2\,\mathrm{d}u = \frac{T^3}{96}$. The error is also orthogonal to ϕ_1 and ϕ_2, since its integral over each half of the interval is zero.

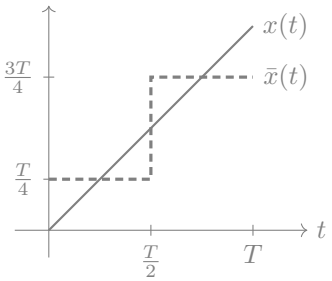


Figure 4.5: The ramp of Exercise 4.2 and its projection onto the basis of Example 4.3.

Exercise 4.3. The first signal has energy T, so the first basis signal is $\psi_1 = s_1/\sqrt{T}$, which is ϕ_1 of Example 4.3. The projection of s_2 onto ψ_1 has coefficient $\langle s_2, \psi_1\rangle = \frac{1}{\sqrt{T}} \cdot \frac{T}{2} = \frac{\sqrt{T}}{2}$, and the error $s_2 - \frac{\sqrt{T}}{2}\psi_1$ equals $1 - \frac{1}{2} = \frac{1}{2}$ on the first half and $0 - \frac{1}{2} = -\frac{1}{2}$ on the second. Its energy is $\frac{1}{4} \cdot T$, so the normalized error is $\frac{2}{\sqrt{T}}$ times it, which equals $\frac{1}{\sqrt{T}}$ on the first half and $-\frac{1}{\sqrt{T}}$ on the second: this is ϕ_2. The Gram-Schmidt procedure applied to the full pulse and the half pulse produces exactly the basis of Example 4.3. The coordinates of s_2 are $\langle s_2, \phi_1\rangle = \frac{\sqrt{T}}{2}$ and $\langle s_2, \phi_2\rangle = \frac{\sqrt{T}}{2}$, and indeed $\frac{\sqrt{T}}{2}(\phi_1 + \phi_2)$ equals 1 on the first half and 0 on the second. Its energy is $\frac{T}{4} + \frac{T}{4} = \frac{T}{2}$, the energy of the half pulse.

Exercise 4.4. The signal u has unit energy, since $\frac{2}{T} \cdot \frac{T}{2} = 1$, and its inner product with ϕ_1 is $\langle u, \phi_1\rangle = \sqrt{\frac{2}{T}} \cdot \frac{1}{\sqrt{T}} \cdot \frac{T}{2} = \frac{1}{\sqrt{2}}$. By linearity, $\langle Y, u\rangle = \pm\sqrt{E}\langle \phi_1, u\rangle + \langle Z, u\rangle = \pm\sqrt{E/2} + Z_u$, where Z_u is Gaussian with variance $\frac{N_0}{2}\|u\|^2 = \frac{N_0}{2}$ by (4.9). The receiver sees a binary antipodal transmission with amplitude $\sqrt{E/2}$ and noise variance $N_0/2$, so by equation (1.23) of Topic 1 the sign detector errs with probability

$$P_\mathrm{e} = \tfrac{1}{2}\operatorname{erfc}\Big(\frac{\sqrt{E/2}}{\sqrt{2}\sqrt{N_0/2}}\Big) = \tfrac{1}{2}\operatorname{erfc}\Big(\sqrt{\frac{E}{2N_0}}\Big), \tag{4.19}$$

the error probability of orthogonal signaling in (4.15): correlating with the half pulse instead of the transmitted

pulse costs 3 dB. In general, a unit-energy u with $\langle u, \phi_1 \rangle = \rho$ gives $\frac{1}{2}\operatorname{erfc}\bigl(\rho\sqrt{E/N_0}\bigr)$, and by Exercise 4.1 $|\rho| \le 1$ with equality only for $u = \pm\phi_1$. The receiver should correlate with the transmitted pulse itself, which is what the correlators of Figure 4.4 do, and Topic 5 calls the filter that does this the matched filter. If u is orthogonal to ϕ_1, then $\rho = 0$, the observation $\langle Y, u \rangle = Z_u$ does not depend on the message, and the receiver can only guess: this is the irrelevance of Theorem 4.12 seen from the wrong side.

Exercise 4.5. In the vector channel (4.11) with $\sigma^2 = N_0/2$, the MAP decoder of Theorem 2.2 of Topic 2 decides $m = 1$ when $q\,W(\boldsymbol{y}|\boldsymbol{x}(1)) > (1-q)\,W(\boldsymbol{y}|\boldsymbol{x}(2))$. Taking logarithms and canceling the common factors of the two Gaussian densities, as in Problem 1.7 of Topic 1, the condition is

$$-\frac{\|\boldsymbol{y} - \boldsymbol{x}(1)\|^2}{2\sigma^2} + \ln q > -\frac{\|\boldsymbol{y} - \boldsymbol{x}(2)\|^2}{2\sigma^2} + \ln(1-q). \tag{4.20}$$

With $\boldsymbol{x}(1) = (\sqrt{E}, 0)$ and $\boldsymbol{x}(2) = (0, \sqrt{E})$, the difference of squared distances is $\|\boldsymbol{y} - \boldsymbol{x}(2)\|^2 - \|\boldsymbol{y} - \boldsymbol{x}(1)\|^2 = (y_2 - \sqrt{E})^2 + y_1^2 - (y_1 - \sqrt{E})^2 - y_2^2 = 2\sqrt{E}\,(y_1 - y_2)$, since the terms in y_1^2, y_2^2 and E cancel. The condition becomes

$$y_1 - y_2 > \frac{\sigma^2}{\sqrt{E}} \ln \frac{1-q}{q} = \frac{N_0}{2\sqrt{E}} \ln \frac{1-q}{q}. \tag{4.21}$$

The decoder looks only at $U = Y_1 - Y_2$, which equals $\pm\sqrt{E} + (Z_1 - Z_2)$ under the two messages, and $Z_1 - Z_2$ is Gaussian with variance $2\sigma^2 = N_0$ because the two projections are independent. This is the binary transmission of Problem 1.7 of Topic 1 with amplitude $A = \sqrt{E}$ and noise variance N_0, whose threshold $\frac{N_0}{2\sqrt{E}} \ln \frac{1-q}{q}$ is the one just found. For $q = \frac{1}{2}$ the threshold is zero, and by equation (1.23) of Topic 1 with threshold $\sqrt{E}$ and variance N_0,

$$P_{\mathrm{e}} = \tfrac{1}{2}\operatorname{erfc}\left(\frac{\sqrt{E}}{\sqrt{2}\sqrt{N_0}}\right) = \tfrac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{E}{2N_0}}\right), \tag{4.22}$$

which is (4.15). The difference $Y_1 - Y_2$ is the projection of Y onto the unit-energy signal $(\phi_1 - \phi_2)/\sqrt{2}$, the direction that joins the two points, scaled by $\sqrt{2}$, and the two waveforms are antipodal along that direction with amplitude $\sqrt{E/2}$.

Exercise 4.6. The shifted pulses $\phi_1(t - (i-1)T)$, $i = 1, \ldots, n$, have unit energy and disjoint supports, so they form an orthonormal set of n signals, one per slot, whose span contains the two waveforms, and in this basis the coordinate vectors are $\boldsymbol{x}(1) = (\sqrt{E}, \ldots, \sqrt{E})$ and $\boldsymbol{x}(2) = -\boldsymbol{x}(1)$. The signal space of the transmitter itself, spanned by s_1 and $s_2 = -s_1$, has dimension one, and we return to it at the end. This is the repetition code of Topic 2 with 2-PAM symbols of amplitude $A = \sqrt{E}$ on the real Gaussian channel with $\sigma^2 = N_0/2$. By Example 2.4(a) of Topic 2, the ML decoder adds the n correlator outputs and decides from the sign of the sum, and by equation (2.15) with $\mathrm{SNR} = A^2/\sigma^2 = 2E/N_0$,

$$P_{\mathrm{e}} = \tfrac{1}{2}\operatorname{erfc}\Bigl(\sqrt{n\,\mathrm{SNR}/2}\Bigr) = \tfrac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{nE}{N_0}}\right). \tag{4.23}$$

A single pulse of amplitude $\pm\sqrt{E/T}$ and duration nT has energy nE and is antipodal signaling in one dimension, with error probability $\frac{1}{2}\operatorname{erfc}\bigl(\sqrt{nE/N_0}\bigr)$ by (4.15), the same value. The two signal sets are in fact the same pair of waveforms, since the sum of the n shifted pulses of amplitude $\sqrt{E/T}$ is the long pulse, and the long pulse divided by $\sqrt{nE}$ is a unit-energy signal, the basis of the one-dimensional signal space, along which both sets are antipodal with amplitude $\sqrt{nE}$. Topic 2 found that repetition with soft decoding gains nothing at equal energy per bit, equation (2.22). In signal space the reason is plain: the error probability depends only on the distance $2\sqrt{nE}$ between the two waveforms, and splitting a pulse into n pieces does not change it.

Exercise 4.7. The inner product is $\langle s_1, s_2 \rangle = A^2 \int_0^T (1 - 2t/T)\,\mathrm{d}t = A^2(T - T) = 0$, so the two waveforms are orthogonal although both occupy the whole interval. The energies are $\|s_1\|^2 = A^2T$ and, with the change of variable $u = 1 - 2t/T$, $\mathrm{d}t = -\frac{T}{2}\,\mathrm{d}u$,

$$\|s_2\|^2 = A^2 \int_0^T \left(1 - \frac{2t}{T}\right)^2 \mathrm{d}t = A^2\,\frac{T}{2} \int_{-1}^{1} u^2\,\mathrm{d}u = \frac{A^2T}{3}. \tag{4.24}$$

By (4.2) with the inner product equal to zero, the squared distance is $\|s_1 - s_2\|^2 = A^2T + \frac{A^2T}{3} = \frac{4A^2T}{3}$, and

by (4.14)

$$P_{\mathrm{e}} = \tfrac{1}{2}\operatorname{erfc}\left(\frac{\sqrt{4A^2T/3}}{2\sqrt{N_0}}\right) = \tfrac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{A^2T}{3N_0}}\right). \tag{4.25}$$

No basis was needed, because distances are computed from energies and inner products alone. The average energy is $E = \frac{1}{2}\big(A^2T + \frac{A^2T}{3}\big) = \frac{2A^2T}{3}$, and antipodal signaling with this energy has $d^2 = 4E = \frac{8A^2T}{3}$, twice the squared distance of the orthogonal pair, so it is 3 dB better, as in Example 4.14.

Exercise 4.8. The M waveforms, normalized, form an orthonormal basis of a space of dimension $n = M$, and the coordinate vector of message m is $\sqrt{E}$ in position m and 0 elsewhere. Any two of them differ in two coordinates by $\sqrt{E}$, so every pair is at squared distance $2E$ and distance $d = \sqrt{2E}$. When message m is sent, an error occurs when some other codeword is nearer to $\boldsymbol{Y}$ than $\boldsymbol{x}(m)$, and the probability of a union of events is at most the sum of their probabilities. By Exercise 2.7 of Topic 2 with $\sigma^2 = N_0/2$, each of the $M-1$ competitors is nearer with probability $\frac{1}{2}\operatorname{erfc}\big(d/(2\sqrt{N_0})\big) = \frac{1}{2}\operatorname{erfc}\big(\sqrt{E/(2N_0)}\big)$, so

$$P_{\mathrm{e}} \le \frac{M-1}{2}\operatorname{erfc}\left(\sqrt{\frac{E}{2N_0}}\right) = \frac{M-1}{2}\operatorname{erfc}\left(\sqrt{\frac{E_{\mathrm{b}}\log_2 M}{2N_0}}\right), \tag{4.26}$$

where $E = E_{\mathrm{b}}\log_2 M$ because each message carries $\log_2 M$ bits. For $M = 2$ the bound is the exact value (4.15). At $E_{\mathrm{b}}/N_0 = 4$ the arguments of erfc are $\sqrt{2}$, 2 and $2\sqrt{2}$ for $M = 2$, 4 and 16, and Table 1.3 of Topic 1 with the supplied value gives

$$P_{\mathrm{e}} \le \tfrac{1}{2}\cdot 4.55\cdot 10^{-2} \approx 2.3\cdot 10^{-2}, \qquad \tfrac{3}{2}\cdot 4.68\cdot 10^{-3} \approx 7.0\cdot 10^{-3}, \qquad \tfrac{15}{2}\cdot 6.3\cdot 10^{-5} \approx 4.7\cdot 10^{-4}. \tag{4.27}$$

At this energy per bit, the bound decreases as M grows. In general, with $\operatorname{erfc}(u) \approx e^{-u^2}/(u\sqrt{\pi})$ for large u and $e^{-a\log_2 M} = M^{-a/\ln 2}$, the bound is asymptotic to $M^{1-E_{\mathrm{b}}/(2N_0\ln 2)}/\big(2\sqrt{\pi E_{\mathrm{b}}\log_2 M/(2N_0)}\big)$. Above $E_{\mathrm{b}}/N_0 = 2\ln 2$, about 1.4 dB, the exponent of M is negative and the bound tends to zero. At equality it still tends to zero, as $1/(2\sqrt{\pi\ln M})$. Below this threshold the bound does not tend to zero. The energy of a message grows like $\log_2 M$ and enters the exponent, while the number of competitors grows linearly. The trend is the opposite of M-PAM in Exercise 1.2 of Topic 1, where each extra bit per symbol costs about 6 dB of symbol SNR at high SNR. The difference is the dimension: M-PAM crowds M points on a line, while orthogonal signaling gives every message its own dimension and uses M dimensions for $\log_2 M$ bits. Topic 6 shows that dimensions cost bandwidth, so orthogonal signaling trades bandwidth for energy, and that this trade has a limit, the capacity of Theorem 3.14 of Topic 3 expressed per second: the minimum energy per bit can approach $N_0\ln 2$ from above as the spectral efficiency tends to zero. This limit is 3 dB below the threshold of the bound.